

# Silicon Slow-Light-Based Modulators and High-Efficiency Optical Transmitters

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**Abstract**— We present CMOS-compatible silicon Mach-Zehnder modulators in which slow-light phase shifters enable a small footprint suitable for dense integration, high-speed at 200 Gb/s and beyond, and low bit-energy consumption below 100 fJ/bit. Such modulators also enable reduced transmitter power consumption through an open-collector configuration.

With the advent of generative AI, the demand for high-throughput data-centre communication has increased dramatically. Optical interconnects based on pluggable transceivers are widely deployed between racks and servers, while co-packaged optics integrating fiber-attached photonic chips near electronic chiplets are under active development. To meet these demands, silicon (Si) photonics must achieve further miniaturization, higher integration density, higher speed, and lower power consumption. Optical transmitters, comprising modulators and driver amplifiers, are key bottlenecks in this regard. Slow light [1] offers a unique route to satisfy these requirements while maintaining full CMOS process compatibility.

Slow light enhances modulation efficiency through an increased group index, enabling substantial reduction of phase-shifter length and drive voltage while maintaining sufficient optical bandwidth. Representative slow-light structures include photonic crystal waveguides [2–4] and grating-assisted rib waveguides [5,6]. By employing a photonic crystal waveguide with a group index exceeding 30, the phase-shifter length of Mach-Zehnder modulators (MZMs) can be reduced to 100–200  $\mu\text{m}$ , resulting in a total modulator footprint of only 0.3  $\text{mm}^2$  including pads and

wiring. This is approximately one order of magnitude smaller than conventional Si MZMs and comparable to microring modulators (MRMs), thereby enabling dense integration for multi-lane and WDM optical transmitters.

At present, the speed of Si MZMs is limited by electro-optic (E–O) phase mismatch and RC time constants. In PCW-based slow-light MZMs, the E–O phase mismatch is compensated by inserting electrical delay lines, improving the cutoff frequency from 18 GHz to 39 GHz [3,4]. However, this value remains approximately half that of standard Si MZMs because the relatively wide photonic-crystal cladding between the waveguide core and electrodes increases parasitic resistance and the RC time constant. To compensate for this limitation, an optical-domain frequency-equalization circuit has been integrated, resulting in a cutoff frequency exceeding 80 GHz. This device achieved 140-Gbaud OOK and 200-Gb/s PAM-4 modulation with only light post-signal processing (Fig. 1) [7]. In grating-assisted rib-waveguide devices, although the group index cannot be increased as flexibly, cutoff frequencies exceeding 100 GHz are achievable, and 200-Gbaud OOK and 400-Gb/s PAM-4 operation have been demonstrated with AI-assisted signal processing [6].

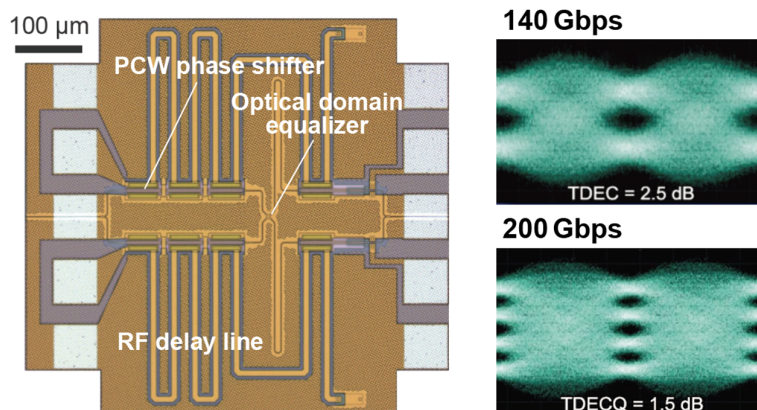


Fig. 1 PCW-based slow-light MZM with optical domain frequency equalizer, which was fabricated using AIST foundry service, and its OOK and PAM-4 operation.

Power consumption in Si MZMs is dominated by dissipation in the termination resistors of the traveling-wave electrodes. By increasing the characteristic impedance and termination resistance to  $50\ \Omega$  through delay-line inductance engineering, modulation power can be significantly reduced. In a PCW-based MZM, the drive voltage has been reduced to  $0.87\ V_{pp}$  and the modulation power to  $3.8\ mW$  at  $64\text{-Gb/s}$  operation, corresponding to  $59\ fJ/bit$  [4]. These values are 1–2 orders of magnitude lower than those of standard Si MZMs and are nearly competitive with MRMs. In MRMs, thermo-optic resonance tuning is required to align the operating point with the optical wavelength, typically consuming roughly an order of magnitude more power than the modulation itself. Therefore, slow-light MZMs offer a clear advantage in overall transmitter energy efficiency when total power consumption is considered.

Power consumption in optical transmitters is dominated by driver amplifiers, but it scales with modulator power. Therefore, low-power slow-light MZMs are already highly effective for reducing total transmitter power. To further reduce this power, a current-driven open-collector driver amplifier is adopted, in which the termination resistor between the driver and modulator is removed and the driver output impedance is doubled. In this configuration, however, impedance matching between the driver and modulator is not inherently

ensured. The two components must therefore be integrated in close proximity, and the amplifier must be carefully designed to suppress unwanted RF reflections. For this purpose, Verilog-A optical device libraries have been developed for co-simulation of photonic and electronic circuits using a de facto standard electronic circuit simulator, enabling amplifier–modulator co-design [8]. Optimized BiCMOS driver chips have been fabricated and hybrid-integrated with a slow-light modulator chip, demonstrating  $64\text{-Gb/s}$  operation with a record-low total power consumption of  $50\ mW$ , corresponding to  $0.78\ pJ/bit$  [9]. By employing state-of-the-art CMOS driver chips and flip-chip integration, this value can be reduced even further [10].

In conclusion, CMOS-compatible and TEC-free slow-light Si MZMs provide a promising route toward compact, high-speed, and energy-efficient optical transmitters for future optical interconnects. They offer significant advantages over MRMs, particularly in co-packaged optics (CPO), where precise thermo-optic control is challenging in the harsh thermal environment adjacent to electronic chiplets.

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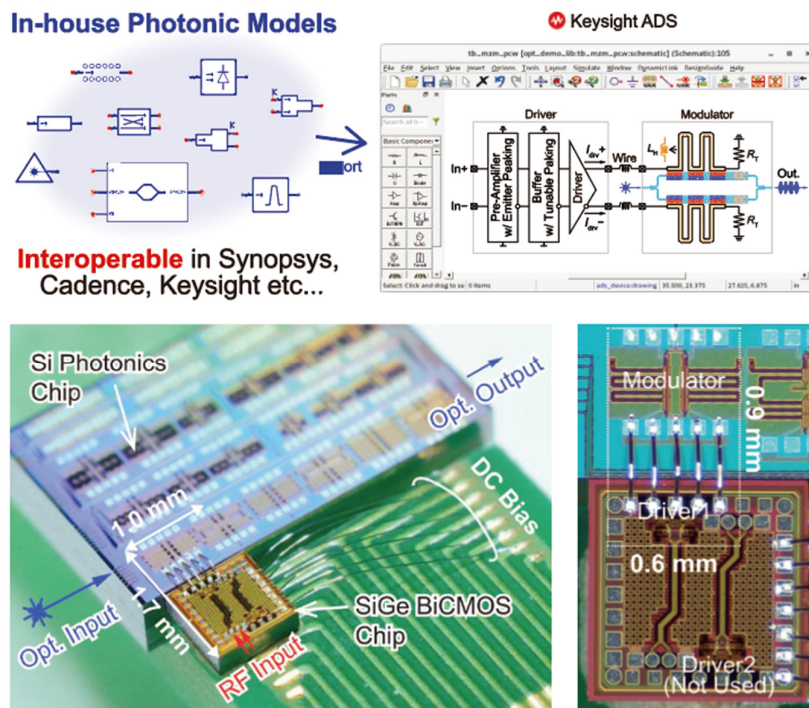


Fig. 2 Photonics-electronics co-simulation environment including in-house photonic model libraries and de facto standard electronic circuit simulator and hybrid integration of open-collector optimized BiCMOS driver and slow-light modulator.

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